

time. Most Linux distros have software manager that lets them search for install and uninstall apps. While this system isn't equipped to handle commercial applications, we now have a number of commercial avenues for Linux software. The Ubuntu Software Centre, for example, is well integrated with the OS, and offers hundreds of free and paid games. Gameolith started as a web-based Linux games store but is not multi-platform, and offers you a copy of the game for all platforms — including Android — with a single purchase.

The biggest Linux game store though is the Steam-like Desura, which supports Windows, Linux and Mac. It may not have the largest collection of games, but its selection of Indie games and mods is excellent. It offers free mods for games it doesn't even sell. They recently made their client open source, so chances are at least some Linux distros will offer it, or come with it installed.

Finally, and most importantly we have Steam. Yes, Steam is coming to Linux, although it isn't there yet. Valve is highly interested in Linux right now, and a Steam client for Linux might be available as soon as this year — you must account for Valve time though and add a few years to that. With Windows 8 coming with an inbuilt store for Metro apps, and Metro looking like the future of Windows, Valve, as a seller of Games for Windows is threatened. They are looking for more open pastures, and Linux seems to be it.

An improving platform

Say what you will, Linux as a platform is getting better, more standardized, more compatible with latest hardware and easier to use. Its rate of adoption is growing, and it already runs on most smartphones. The Linux Standard Base



Ubuntu Software Center

TORVALDS: "NVIDIA: F*** YOU"

Famous words as spoken by Linus Torvalds in a recent question answer session following a presentation. The creator of the Linux kernel made it known in no uncertain terms that NVIDIA is "the single worst company" that they have ever dealt with. Now those who use Linux and have an NVIDIA card will know that the NVIDIA-provided drivers are superb, so why the strong words?

Linux is a different beast than Windows. Linux users rarely have to worry about drivers, everything just works. This is why Live CDs of Linux work so well and are so popular. Graphics drivers however are something Linux users still have to worry about, and they are rather essential. For this kind of core support the hardware specifications need to be known by kernel developers. NVIDIA like many other companies keeps these specifications secret to "protect" their technology. This leaves Linux developers with no choice but to reverse engineer the drivers — in a project called nouveau — to create an open source version, a time-consuming and hard process. As a result, the performance of Linux on NVIDIA hardware is severely lacking. You might get your 3D desktop, but few 3D games will not work at all unless the closed source drivers are installed and those are not included with the distro by default. Also, NVIDIA Optimus, a technology for switching between a dedicated and integrated graphics cards automatically to save energy, is not supported on Linux even by NVIDIA's own drivers.

In contrast, AMD have released some of their specifications and actively develop an open source driver, as do Intel. Despite having better drivers than AMD for their hardware, NVIDIA are not good open-source citizens. This fact has recently even cost them a rather large order (millions) of graphics cards.

(LSB) project aims to develop a standard that all Linux distros follow. Few do in its entirety, but the situation is nonetheless improving.

Another important factor in this equation is Wine, a software that makes it possible to run Windows applications on Linux directly. It acts as a translator between the Windows application and the Linux system underneath. Wine is steadily improving and can now run popular games such as *World of Warcraft*, *StarCraft II*, *Team Fortress 2*, *Half-Life 2*, and even new titles such as *Mass Effect 3* and *Diablo III*.

The multi-platform revolution

Just a decade ago, you would have heard from a developer that supporting more than one platform was out of their reach, scope, budget or some other noun for an excuse. Today, nearly every popular application or service is multi-platform, and there are more primary platforms than ever. You have Windows, OS X, Linux, Android, and iOS. The lesser used, but still major platforms are Windows Phone, Blackberry and Symbian. Yet there are some applications that are available on all of these platforms. For games there are also Xbox 360, PlayStation 3 and the Wii.

Cross-platform development encourages good development practises, and ensures people avoid depending on the underlying OS too much. Once you have achieved that, reaching other platforms is much smoother. Look at how many of the recent games are available for OS X now; most Valve games, *Diablo III*, *Assassins Creed II*, *Deus Ex: Human Revolutions*, *Batman: Arkham City*, *Dragon Age* games and many more. There is also matter of DirectX and OpenGL. Being a Microsoft technology DirectX is only available on Windows and Xbox while OpenGL is an open standard and runs everywhere. A subset of OpenGL is also used in WebGL and on all mobile platforms — other than Windows Phone, of course — so it is something developers have to be familiar with, or at least target now. If a game is running on OS X, Linux is only a stone's throw away.

A number of new game engines now pride themselves on their cross-platform abilities. Recently Linux support was announced for the popular Unity game engine in the next v4; they credited this to the Humble Bundles and Ubuntu Software Centre. Moai is another example of an engine that is open source and supports running on iOS, Android, Windows, Linux, OS X, and the Chrome browser. Unigine is a high quality engine that supports Windows, Linux, OS X, Android, iOS, and PS3. There are many more that are more selective, but still multi-platform and could come to Linux as they already support OS X. Gaming on Linux might seem like a trivial thing to worry about, but it is essential to the ecosystem as a whole. The lack of games is often cited as one of the things holding Linux back. Having one of the most popular and fun uses of computers available to Linux will only serve to improve the adoption of Linux. 